5.3 Isomorphisms

Homomorphisms can be one-to-one or onto. When a homomorphism is both one-to-one and onto it is a bijection and bijective homomorphisms play a special role in group theory.

Definition 5.3.1. If $\theta : G \to H$ is a bijective homomorphism then we call θ an *isomorphism* and say that G and H are *isomorphic* and write $G \cong H$.

For example, $\mathrm{Sym}(\{a, b, c\}) \cong \mathrm{Sym}(\{1, 2, 3\})$.

Isomorphic groups are the same group except that they may use different names for their elements. We sometimes say things like "up to isomorphism there is only one group of order 7" to indicate that we consider isomorphic groups to be the same.

For example, up to isomorphism there is precisely one group of order 7. It is C_7. But there are in fact infinitely many groups of order 7 — they are just all isomorphic. Here are some of them:

$$\langle(1\ 2\ 3\ 4\ 5\ 6\ 7)\rangle \;\cong\; \langle(a\ b\ c\ d\ e\ f\ g)\rangle \;\cong\; \langle(11\ 12\ 13\ 14\ 15\ 16\ 17)\rangle \;\cong\; \cdots$$

Example 5.3.2. Here are some examples of isomorphisms.

(i) Let $n \in \mathbb{N}$. Recall $C_n = \langle \rho \rangle$, where ρ is the n-cycle $(1\ 2\ \cdots\ n)$, and $\mathbb{Z}_n = \langle [1]_n \rangle$.
Every element in C_n looks like ρ^m for some m, and every element in $\mathbb{Z}_n$ looks like $[1]_n \oplus \cdots \oplus [1]_n$ (m times).
We have $C_n \cong \mathbb{Z}_n$ via the isomorphism $\varphi : C_n \to \mathbb{Z}_n$ where $\varphi(\rho^m) = [1]_n \oplus \cdots \oplus [1]_n$.
Exercise: Check yourself that φ is an isomorphism (check it is a homomorphism, and is onto and one-to-one). Once you've done this, you can check your answer in the handout where I go through the argument.

(ii) Here is an example you will be familiar with. The positive real numbers under multiplication form a group $(\mathbb{R}_{>0}, \times)$, and the real numbers under addition form a group $(\mathbb{R}, +)$.
The natural log function $\ln : \mathbb{R}_{>0} \to \mathbb{R}$ is an isomorphism from $(\mathbb{R}_{>0}, \times)$ to $(\mathbb{R}, +)$. You can prove this as an easy exercise.

> Hint: The fact that ln is onto and one-to-one is clear if you plot the graph of $y = \ln x$, for $x \in \mathbb{R}_{>0}$. Moreover, $\ln(x \times y) = \ln(x) + \ln(y)$, so ln is a homomorphism.

Non-Example 5.3.3. Here are some non-isomorphic groups.

Suppose G and H are finite groups, with $|H| < |G|$. Then $G \ncong H$.
Proof: If there is a bijection between two finite sets, then the sets must have the same number of elements (you will have proved this in your first year). Hence, if there exists an isomorphism θ from G to H, then (because θ is also a bijection from G to H) the two groups must have the same number of elements, which they do not.

We conclude by showing that for any prime p, there is (up to isomorphism) only one group of order p.

Theorem 5.3.4. *Let G be a group and p be a prime. If $|G| = p$, then $G \cong C_p$.*

Proof. We first prove that G is cyclic. Choose an nonidentity element $g \in G$ and let $H = \langle g \rangle$. By Lagrange's Theorem, $|H|$ divides $|G| = p$. Since p is prime, we have $|H| = 1$ or $|H| = p$.
By Proposition 2.2.6, we know that $|H| = 1$ if and only if $H = \langle e_G \rangle$. This cannot be the case, because $g \neq e_G$. Hence $|H| = p$. Therefore $|H| = |G|$, and so again by Proposition 2.2.6 we have $G = H$. Thus, G is cyclic and generated by g. Hence

$$G = \{e_G, g, g^2, \ldots, g^{p-1}\}.$$

Recall that $C_p = \langle \rho \rangle$, where ρ is the cycle $(1\ 2\ \ldots\ p)$, and so we have

$$C_p = \{e, \rho, \rho^2, \ldots, \rho^{p-1}\}.$$

Let $\varphi : G \to C_p$ be given by

$$\varphi(g^n) = \rho^n \qquad \text{for all } 0 \le n < p.$$

It is easy to see this function is one-to-one and onto. Hence it is a bijection. Moreover, it is a homomorphism, because $\varphi(g^n g^m) = \varphi(g^{n+m}) = \rho^{n+m} = \rho^n \rho^m = \varphi(g^n)\varphi(g^m)$. $\square$

5.3.1 Handout for Section 5.3

Proof that the function φ from Example 5.3.2(i) is an isomorphism. Let ρ be the permutation $(1\ 2\ \cdots\ n)$. We know that $C_n = \langle\rho\rangle$ by definition and $\mathbb{Z}_n = \langle[1]_n\rangle$ by Question 2.5.6. Recall that the operation in C_n has no symbol, but the operation in $\mathbb{Z}_n$ is $\oplus$.

(Be careful! $([1]_n)^m = [1]_m \oplus \cdots \oplus [1]_n = [m]_n$.)

Let $\varphi : C_n \to \mathbb{Z}_n$ be the map $\varphi(\rho^m) = ([1]_n)^m$ for all $m \in \mathbb{Z}$. First, we show φ is a homomorphism, then we show it is onto, and finally we show it is one-to-one.

[Homomorphism] Fix two elements in C_n, say ρ^r and ρ^s. Then $\varphi(\rho^r\rho^s) = \varphi(\rho^{r+s}) = ([1]_n)^{r+s} = [r+s]_n = [r]_n \oplus [s]_n = \varphi(\rho^r) \oplus \varphi(\rho^s)$.

[Onto] Choose any element in $\mathbb{Z}_n$, say $[m]_n$ for some $m \in \{0, 1, \ldots, n-1\}$ and note that $\varphi(\rho^m) = [m]_n$. Thus φ is onto.

[One-to-one] If $\varphi(\rho^r) = \varphi(\rho^s)$ then:

(multiplying both sides by $\varphi(\rho^s)^{-1} = \varphi(\rho^{-s})$)

we have $\varphi(\rho^{-s}) \oplus \varphi(\rho^r) = \varphi(\rho^{-s}) \oplus \varphi(\rho^s)$. Hence $\varphi(\rho^{-s+r}) = \varphi(\rho^{-s+s})$, because φ is a homomorphism. Thus, $[-s+r]_n = [-s+s]_n = [0]_n$. This implies that $\rho^{-s+r} = \rho^0 = e$. Multiplying both sides by ρ^s gives $\rho^r = \rho^s$. In other words, $\varphi(\rho^r) = \varphi(\rho^s)$ implies that $\rho^r = \rho^s$. Thus φ is one-to-one. $\square$